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RESEARCH

Integration of parameterized quantum circuits within classical neural network for financial time-series prediction

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Abstract
The forecasting of financial defaults plays a pivotal role in augmenting risk management methodologies and ensuring the resilience of financial ecosystems. This study presents the seamless integration of classical and quantum computing methodologies for developing a neural network for financial default prediction on a time-series dataset. A hybrid neural network architecture also known as a dressed quantum circuit, is proposed to integrate classical and quantum layers for financial default prediction. The state-of-the-art dressed quantum circuit, incorporates rotations and multiqubit controlled gates meticulously tailored to navigate the intricacies of complex, large-scale, and high-dimensional time-series datasets. The contributions of the study are multifaceted. Firstly, the study presents a distinctive framework for the integration of quantum computing with classical machine learning, heralding a promising pathway for addressing high complexity and dimensionality challenges. Secondly, the study demonstrates the scalability of the model's configuration through variations in the number of qubits and quantum circuit depth and optimizes the model's performance through the application of diverse optimization algorithms. Thirdly, the model's enhanced efficiency is evidenced by a significant reduction in training time compared to state-of-the-art classical gradient-boosting algorithms. The hybrid model has the potential and adaptability to offer robust solutions across various real-world problems.

Keywords Machine learning · Quantum computing · Neural networks · Parameterized quantum circuits

1 Introduction

The advent of quantum computing has ushered in a new era of computational capabilities, promising to address some of the most challenging problems across various domains. In the realm of data analytics, the limitations of traditional machine learning models for handling complex, high-dimensional data have become increasingly apparent. Quantum computing represents a significant leap forward from classical computing, offering a new paradigm for processing information. Unlike classical computing, which relies on bits that exist in a state of 0 or 1, quantum computing uses qubits, which can exist in multiple states simultaneously due to superposition and other quantum phenomena, such as entanglement, that provide quantum computers with capabilities far beyond what is achievable with classical technology (Feynman 1982; Gupta and Nene 2021). A fundamental advantage of quantum computing is its inherent parallelism due to superposition. A quantum system of n qubits can represent and process 2^n possible states simultaneously (Grover 1997). This contrasts sharply with classical systems, which would need to process each of these states sequentially. The ability to handle multiple computations at once opens up new possibilities for solving complex problems more efficiently. Another important property exhibited is entanglement, which allows qubits to be correlated in such a way that the state of one or more cannot be described independently of the state of another, even when the qubits are separated by large distances (Feynman 1982). The properties of superposition and entanglement can be exploited in algorithms to solve certain types of problems much more quickly than classical algorithms, particularly those requiring the exploration of a vast number of possible solutions

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or configurations (Grover 1997; Khurana and Nene 2023a; Rønnow et al 2014). Quantum computers can simulate quantum systems efficiently, a task that is intractably complex for classical computers as system size grows. This capability is particularly advantageous for understanding molecular structures, reaction dynamics, and material properties, potentially accelerating the discovery of new materials, drugs, and chemical processes (Cirac 2021; Cao et al 2019). Many problems across logistics, finance, and engineering can be framed as optimization problems, which can be efficiently solved using quantum computing (Jones 2021; Mathur 2019). Quantum computing offers novel approaches to machine learning by including faster matrix operations, quantum state classification, and the development of neural networks, which could potentially revolutionize Artificial Intelligence (AI) (Khurana and Nene 2023b; Jhanwar and Nene 2021). The study is motivated by the possible benefits of integrating quantum computing with classical machine learning techniques to form a hybrid machine learning model, aiming to harness the unique advantages of quantum computation to enhance predictive analytics.

1.1 Enhancing machine learning with quantum computing: overcoming the limitations of classical computing

Classical computing has been the backbone of technological advancements and has propelled innovations across various fields, from healthcare and engineering to artificial intelligence and beyond. However, as we push the boundaries of data volume, computational complexity, and the need for more sophisticated algorithms, classical computing faces significant limitations. Classical systems face difficulties solving problems that require exploring an exponential solution space within reasonable time frames or with desired accuracy. This bottleneck has led to a search for alternative computing paradigms capable of transcending these limitations.

The intersection of quantum computing and machine learning presents unprecedented opportunities to accelerate learning algorithms, optimize neural networks, and process information with quantum-enhanced efficiency (Farhi and Neven 2018). Quantum algorithms can potentially analyze vast datasets more swiftly and accurately than classical algorithms, opening new frontiers in machine learning research and applications (Rønnow et al 2014; Cirac 2021; Farhi and Neven 2018). This synergy between quantum computing and machine learning could lead to more intelligent systems capable of complex decision-making, pattern recognition, and predictive analytics far beyond their current capabilities. The motivation for advancing quantum computing is driven by the quest to transcend the constraints of classical computing, harness the power of quantum mechanics for computation, and open new avenues for solving some of the most intricate problems.

1.2 Objective

The objective of this study is to design and investigate the effectiveness of a Hybrid Neural Network (HNN) comprising both classical and quantum layers in the specific context of predicting credit card default risk on a time-series financial dataset. The quantum layer's ability to process information in superposition and entanglement allows for a more nuanced understanding of complex data like financial data, leading to better expressibility and faster training compared to entirely classical models. Further, it also aims to build an HNN that generalizes effectively across varied datasets and possesses the scalability necessary to accommodate increasing data size and complexity, thereby ensuring its applicability and efficacy in the face of evolving computational demands.

1.3 Contributions

This paper contributes to the burgeoning field of quantum machine learning (QML) applications in financial risk assessment by demonstrating the potential of employing parametrized quantum circuits (PQCs) in a classical neural network for credit default prediction. The study addresses both the theoretical and practical aspects of quantum-enhanced predictive analytics. The contributions of this study are manifold:

- **Innovative integration framework:**The study introduces a framework that seamlessly combines quantum computing with classical machine learning techniques. This integration marks a significant step forward in tackling challenges associated with high-dimensional data and computational complexity, offering a novel approach to problems previously deemed complex.
- **Scalability and optimization:** Through meticulous experimentation with the model's configuration, including variations in qubit count and quantum circuit depth, the study embarks on a comprehensive exploration of scalability. Further, the study distinguishes itself by optimizing model performance through the application of a spectrum of optimization algorithms, showcasing the model's flexibility and adaptability to different computational environments.
- **Training efficiency:** A noteworthy outcome of this research is the reduction in training time when juxtaposed with contemporary classical gradient-boosting algorithms. This efficiency gain not only signifies an improvement in computational resource utilization but also enhances the practicality of deploying such hybrid models in time-sensitive applications.

- **Generalizability and applicability:** The hybrid model's ability to generalize on unseen data and its proven effectiveness in navigating complex real-world scenarios underscore its robustness and versatility. This adaptability ensures the model's relevance and utility in addressing a wide range of multifaceted problems.

This novel approach exemplifies the broader potential for quantum computing to bring about a revolution in predictive analytics, offering insights into its applicability and effectiveness in real-world scenarios.

1.4 Outline

This paper has been organized into sections, in which Section 2 delves into existing literature, wherein quantum-enhanced machine learning approaches utilize PQCs, highlighting advancements, applications, challenges, and gaps that this work aims to address. Section 3 discusses the methodology, providing insights into the pre-processing of the dataset, the experimental setup, network architecture, and quantum circuit design, and the selection of optimization algorithms. Section 4 presents the results achieved through a thorough analysis of the model's performance with varying parameters. Section 5 discusses and reflects on the findings, discussing the implications of integrating quantum computing with machine learning, the scalability of quantum approaches, their practical applicability in real-world scenarios, and potential avenues for future research, followed by a conclusion in Section 6.

2 Related work

HNNs utilizing PQCs represent a novel intersection of quantum computing and artificial intelligence. The review outlines the theoretical foundations, recent advancements, applications, challenges, and future directions in the development and application of QNNs utilizing PQCs. This section reviews the existing literature across these domains, identifying the advancements made and the challenges that the study intends to address. HNNs that utilize PQCs represent an innovative approach that combines classical deep learning techniques with quantum computing principles to enhance machine learning tasks (Benedetti et al 2019; Mari et al 2020; Emmanoulopoulos and Dimoska 2022). These networks leverage the power of quantum computation, particularly its ability to efficiently process complex, high-dimensional data spaces using quantum superposition and entanglement (Cerezo et al 2021; Chen et al 2021; Hong et al 2023). PQCs serve as the quantum layers within the network, where parameters are tuned similarly to weights in classical neural networks, offering a novel way to encode and process information. The combined approach is aimed at leveraging the advantages of both classical and quantum systems to potentially address intricate optimization problems that classical algorithms struggle with due to computational constraints. PQCs, also referred to as variational quantum circuits, are at the forefront of the fusion between quantum computing and machine learning (Hubregtsen et al 2021). These circuits are characterized by their use of quantum gates, whose parameters can be tuned or optimized, much like the weights in classical neural networks. This adaptability allows PQCs to gain insights from data, making them a powerful means for machine learning tasks. PQCs utilize the principles of superposition and entanglement of qubits to perform computations (Benedetti et al 2019). This allows PQCs to model complex, nonlinear relationships within the data. PQCs have been explored for a variety of QML tasks, highlighting the versatility of PQCs and their potential to enhance machine learning capabilities through quantum computation (Cerezo et al 2021). These PQCs can form an essential building block in the optimization of machine learning models. Recent studies have further explored the theoretical foundations of PQCs, emphasizing their potential for enhancing machine learning models through quantum computation.

2.1 HNNs utilizing PQC for machine learning

In Hubregtsen et al (2021) conducted an evaluation of PQCs, addressing the challenge of designing circuits that optimally represent the solution space with minimal circuit depth and parameter count. The study reveals that sequences of controlled X-rotation gates outperform those of controlled Z-rotation gates in improving both expressibility and entangling capability. The investigation explored the phenomenon of expressibility "saturation" with increasing circuit depth. This work underscores the importance of circuit design in leveraging the quantum advantage for machine learning tasks. The study was carried out on datasets comprising a total of 1500 data points (Hubregtsen et al 2021).

In Du et al (2020) highlighted the expressive power of PQCs by explaining the theoretical foundations and proving that even simple PQCs can outperform classical neural networks. This suggests a significant theoretical advantage of quantum circuits in machine learning, particularly for machine learning tasks. However, empirical demonstrations on machine learning tasks were not conducted to substantiate any definitive quantum advantage.

In Xia and Kais (2020) presented an HNN designed to compute the ground state energy of molecules, marking a significant advancement in quantum chemistry research. This innovative approach utilized PQCs integrated with classical neural network structures, introducing non-linearity through quantum measurements to perform calculations.

Numerical simulations demonstrated the model's accuracy in approximating ground state energies, highlighting the potential for applying hybrid quantum-classical systems to complex molecular energy surface generation. This work showcases the feasibility of combining quantum and classical computing techniques for quantum chemistry problems. However, multidimensional features can be utilized to demonstrate the quantum advantage over classical machine learning.

In Zhao and Wang (2021) reviewed the progress and current status of QNNs by systematically categorizing QNNs into different models based on their architecture and operational principles, including variational quantum circuits and quantum versions of classical neural networks. It delves into the methodologies employed in the construction and training of QNNs, highlighting the unique challenges such as quantum data encoding, gradient computation in quantum systems, a lack of quantum hardware making it difficult to fully leverage the quantum advantage inherent in quantum neural networks on low-dimensional datasets, and problems with smaller sample sizes.

In Verdon et al (2019) explored a novel framework that integrates QNNs with classical neural networks to enhance the learning capabilities of quantum systems. The paper introduced a meta-learning (Verdon et al 2019) approach where a classical neural network is employed to optimize the PQCs that constitute the QNNs. This hybrid model leverages the strengths of both computing paradigms, aiming to address the challenges associated with the training of QNNs including the barren plateau phenomenon and the effective conversion of classical data into quantum states. However, the challenge persists in the optimization of neural networks, necessitating meticulous fine-tuning of the parameters to achieve optimal performance.

In Senokosov et al (2023) proposed two hybrid quantum-classical models for image classification on the MNIST dataset by integrating quantum circuits into neural networks to enhance performance in image recognition tasks. This study tackled the computational hurdles inherent in image classification, a task exacerbated by the increasing volume of visual data. Neural networks with quantum circuits circumvented the limitations posed by the current infeasibility of circuits with a large number of qubits. This approach underscored the model's generalizability and the quantum layers' efficiency in feature extraction. The paper emphasizes on further development of optimization techniques, underscoring the imperative for advanced strategies in the enhancement of quantum model performance.

In McClean et al (2018) elucidated how the high-dimensional landscape of QNN parameters can lead to regions where the gradient of the cost function becomes exponentially small as the system size increases. This phenomenon significantly hampers the efficiency of gradient-based optimization methods, making it difficult to train QNNs as they scale. Through theoretical analysis and numerical simulations, it demonstrated that barren plateaus are a common occurrence in randomly initialized quantum circuits, especially those with global cost functions. The study highlighted the importance of careful circuit design and initialization strategies to mitigate the impact of barren plateaus, offering insights that are crucial for the advancement of QNNs.

2.2 Observations and challenges

The literature review illuminates a spectrum of critical challenges to foster progress in HNNs, encompassing the scalability of the model (Zaman et al 2024), the imperative for seamless integration between computing paradigms (Arthur and Date 2022), constraints imposed by the prevailing Noisy Intermediate-Scale Quantum (NISQ) era (Preskill 2018), and the capacity to address intricate real-world problems (Hubregtsen et al 2021).

- **Integration of classical and quantum computing:** A primary challenge lies in harmonizing classical machine learning algorithms with quantum computing principles. This integration necessitates novel methodologies that can benefit from the unique computational advantages of both paradigms while retaining the robustness and versatility of classical machine learning techniques.
- **Design of quantum circuits:** The current NISQ era presents significant constraints in terms of quantum hardware capabilities (Sim et al 2019; Preskill 2018). Designing quantum circuits that are compatible with NISQ technologies and are scalable for future advancements is paramount. These circuits must be adept at handling the noise and error rates characteristic of present-day quantum computers, ensuring reliable performance as quantum technology evolves.
- **Experimentation with complex datasets:** There is a pressing need for comprehensive experimentation involving large, high-dimensional, and complex datasets. Such experimentation is crucial for assessing the efficacy of hybrid models in capturing intricate patterns and relationships inherent in real-world data, thereby pushing the boundaries of what is computationally feasible.
- **Practical implementation on real-world problems:** Bridging the gap between theoretical models and practical applications remains a significant hurdle. The ability to implement hybrid quantum-classical neural networks on real-world problems validates their practical utility as

well as opens up avenues for solving problems that were previously intractable with classical computing alone.

The literature highlights the critical need for innovative solutions to these challenges to maximize the capabilities of hybrid neural networks. Addressing these challenges will contribute to advancing the field of quantum machine learning by providing groundbreaking applications in many fields.

3 Methodology

To address the identified gaps and challenges in the integration of classical machine learning methods with quantum computing and the design and practical application of HNNs, the methodology is structured around several core objectives. These objectives are aimed at overcoming the limitations of current quantum hardware, optimizing quantum circuits for scalability and efficiency, and facilitating the practical implementation of these systems in solving intricate, real-world problems. A comprehensive approach has been adopted to develop, train, and evaluate a hybrid neural network for the task of predicting credit card default risk. This section delves into the process of data preparation, feature selection strategies, the proposed architecture of the hybrid neural network, the experimental setup, and the training and evaluation of the model.

3.1 Dataset

The study utilized a dataset procured from the Kaggle platform, an open-source repository, to estimate the probability of a customer failing on their credit card payment in the future (Amex 2024a). The dataset comprises a total of thirteen monthly statements for each customer, encompassing a wide range of features and representing various aspects of customer behavior and account information. The dataset comprises 5,531,451 records, each corresponding to a monthly statement for individual customers over a fixed period for 458,913 unique customers. The dataset includes several categories of features, each prefixed to indicate their nature.D_, Delinquency variables, indicating past due payment behaviors, S_,Spend variables, representing the amount spent on the account,P_, Payment variables, detailing payment amounts made by the customer, B_, Balance variables, showing the account balance, R_, Risk variables, calculated based on the account's risk profile. Each record is associated with a unique "customer_ID" and a statement date "S_2," alongside the anonymized features (Amex 2024a).

3.2 Data preparation

The data preparation phase began with an Exploratory Data Analysis (EDA) to understand the dataset's characteristics, including the distribution of values, the presence of missing data, and potential correlations among features. Normalization of features is carried out to accommodate the sensitivity of the model to the scale of the data. Missing values are addressed either by imputation or by omitting features with a high percentage of missing data, depending on the nature and importance of the feature. Categorical variables are encoded using one-hot encoding, transforming them into a format that can be processed by the neural network.

3.3 Feature selection

The original dataset consists of a large number of features. Feature selection is conducted to determine the most significant features for forecasting credit card default risk. Features highly correlated with the target variable are prioritized, while features with little to no correlation are considered for removal, leading to the selection of the top 64 features by utilizing the Chi score method.

3.4 Model architecture

The hybrid neural network specifically developed to predict the credit card default on the time-series data involved the integration of classical and quantum computing paradigms to leverage their respective strengths in processing and analyzing data. The architecture is structured as follows, as depicted in Fig. 1.

- **Classical input Layer:** This layer acts as the initial point of contact for the input data, performing linear transformations followed by non-linear activation. This layer preprocesses the input data and transforms it into a suitable form for data encoding by the quantum circuit. The transformation is mathematically represented as:

$$f(x) = \sigma(Wx + b) \tag{1}$$

  where W is the weight matrix, x is the input vector, b is a bias vector, and σ is a non-linear sigmoid activation function.
- **Quantum layer:** At the core of the HNN is the quantum layer, implemented using PQCs. This layer utilizes a set of quantum gates with adjustable parameters analogous to weights in classical neural networks to perform complex transformations on the data encoded into quantum states. Here, rotation gates are initialized with random angles within the quantum circuit. This process encodes

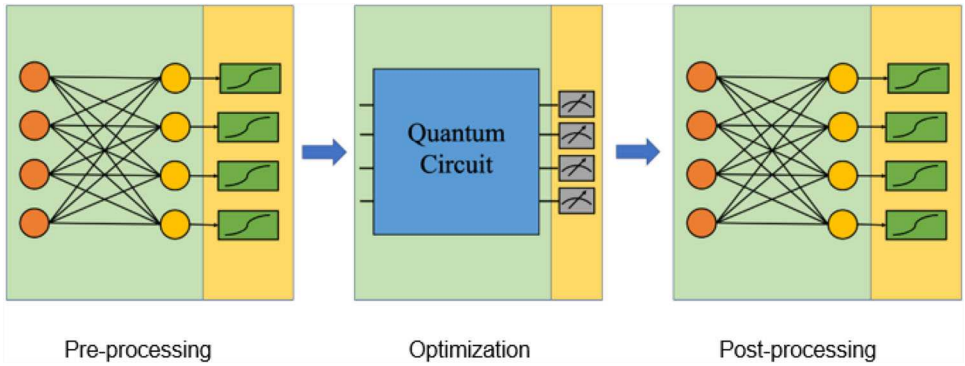


Fig. 1 Network architecture for hybrid neural network

the preprocessed classical data into the quantum state and performs quantum computations based on the circuit configuration.

Figure 2 depicts the PQC developed and used in the architecture for conducting experiments, initialized with the six qubits. The RY gate is used to rotate around the y-axis on the Bloch sphere by a certain angle, as rotations are fundamental for creating superpositions and entangled states in quantum computing. RY gate is followed by controlled-NOT (CNOT) gates; the gate flips the state of the target qubit if the control qubits are in the state $|1\rangle$. The multi-qubit controlled unitary CNOT gates applied in this architecture entangle each qubit with others in order to capture the correlation among the features. The circuit has been experimented with, comprising the sequence of operations RY gate followed by multi-qubit controlled unitary CNOT gates, as one layer across all qubits and can be applied multiple times. The rotation angles derived from the quantum circuit are adjusted to minimize the loss function.

- **Classical output layer:** Following the quantum layer, another classical layer interprets the quantum processing results. For binary classification tasks like credit default prediction, a single output neuron with a sigmoid activation function has been used to produce a probability score indicating the likelihood of default. The transformation can be represented as:

$$g(y) = \sigma(W'y + b') \tag{2}$$

where W' is the weight vector, y is the input vector from the quantum layer's output, b' is a bias term, and σ is a non-linear sigmoid activation function.

In the hybrid neural network architecture, the first classical layer is responsible for pre-processing the input data and transforming it into a suitable form for quantum encoding. The quantum circuit layer then performs quantum computations, encoding and manipulating the data in a quantum state. Finally, the last classical layer processes the quantum layer's output, transforming it back into a classical form for interpretation or decision-making. This hybrid approach allows for the exploitation of quantum computing's advantages for specific computational tasks while maintaining the versatility and accessibility of classical data processing techniques.

4 Experiments and results

A systematic experimental approach is adopted to investigate the influence of various architectural and optimization parameters on the model's performance. This investigation

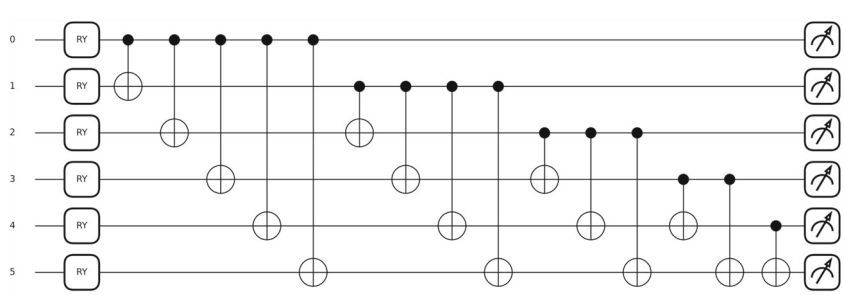


Fig. 2 Schematic representation of parameterized quantum circuit

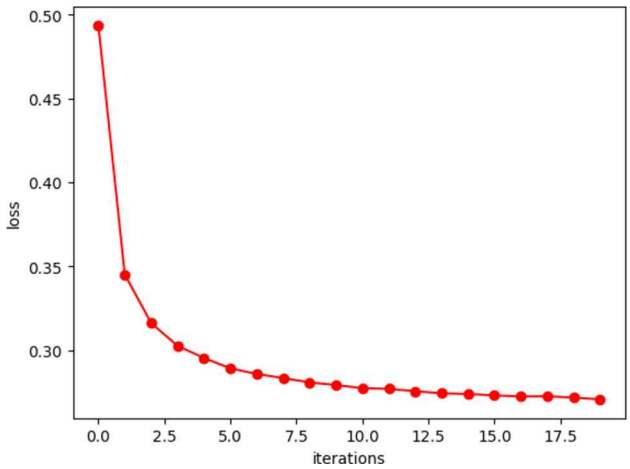


Fig. 3 Loss function for five-qubit PQC architecture

Table 1 Performance metrics for the training and test datasets for five-qubit PQC architecture

Dataset	Metric	Class 0	Class 1	Macro avg	Weighted avg
Training	Precision	0.91	0.79	0.85	0.88
	Recall	0.93	0.75	0.84	0.88
	F1-score	0.92	0.77	0.85	0.88
	Accuracy				0.88
Test	Precision	0.91	0.76	0.84	0.88
	Recall	0.93	0.77	0.85	0.88
	F1-score	0.92	0.77	0.85	0.88
	Accuracy				0.88

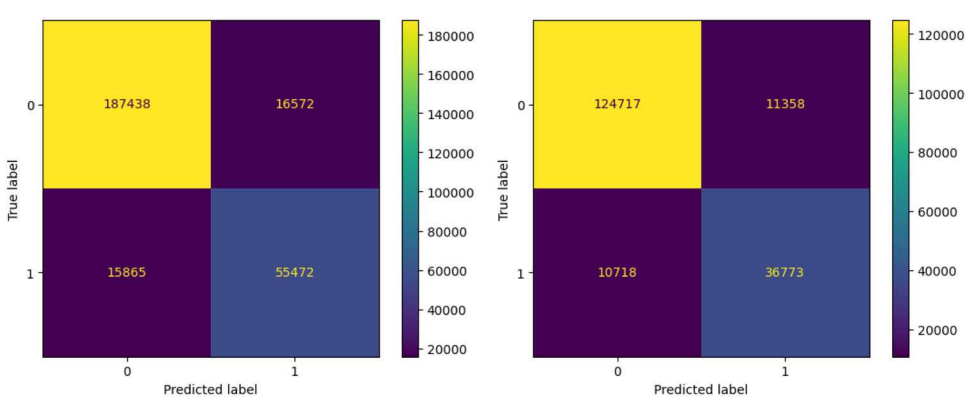
is structured around a series of experiments designed to explore the multidimensional parameter space that defines the model's architecture and operational dynamics. Specifically, the experiments are centered on three principal axes of variation: the number of qubits constituting the PQC, the selection of optimization algorithms, and the depth of the quantum circuit as determined by the number of quantum layers. These key hyperparameters are systematically varied to identify configurations that offer optimal performance and also to assess the scalability of the model. This exploration provided valuable guidelines for the design and implementation of HNN models tailored to specific computational tasks and datasets.

4.1 Experimental setup

- **Hardware** Google Colab Pro has been utilized to conduct experiments by leveraging the high computational power of a CPU with 51 GB of RAM which is required for a huge amount of training data.
- **Software** For the execution of the machine learning model presented in this study, the computational environment utilized included Python version 3.12.2, along with essential libraries including NumPy version 1.26.4, Pandas version2.0.0, Matplotlib version 3.6.2, Scikit-learn version 1.1.3, and JAX version 0.4.1. These software requirements are critical for handling data preprocessing, model implementation, optimization, and evaluation, ensuring compatibility and efficiency in replicating the research findings.

4.2 Experimental configurations

The core of the investigation is centered around a hybrid neural network architecture, which integrates classical data pre-processing and post-processing layers with an intermediary quantum circuit. This hybrid model is subjected to



(a) Confusion Matrix for training dataset (b) Confusion Matrix for test dataset

Fig. 4 Confusion matrix for five-qubit PQC architecture

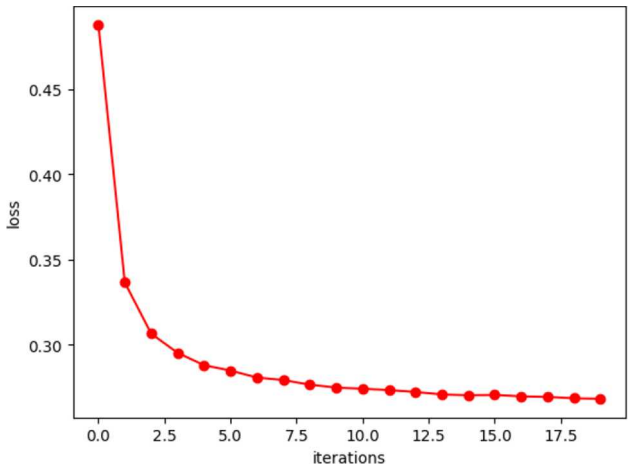


Fig. 5 Loss function for six-qubit PQC architecture

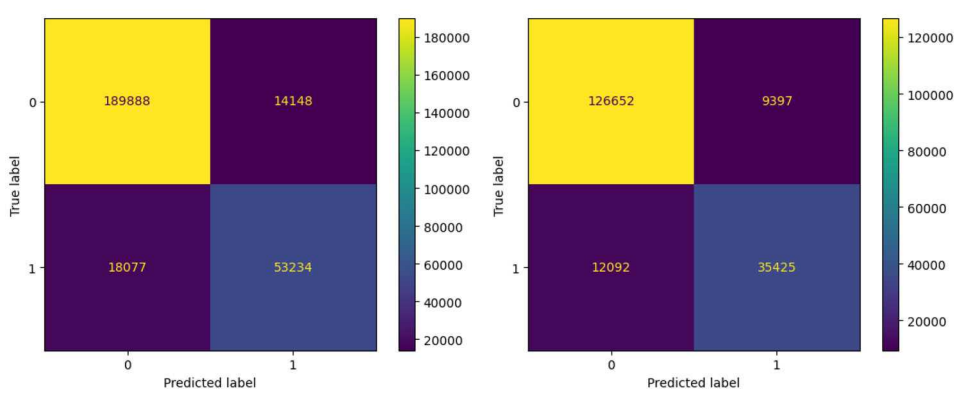
Table 2 Performance metrics for the training and test datasets

Dataset	Metric	Class 0	Class 1	Macro avg	Weighted avg
Training	Precision	0.91	0.79	0.85	0.88
	Recall	0.93	0.75	0.84	0.88
	F1-score	0.92	0.77	0.84	0.88
	Accuracy				0.88
Test	Precision	0.91	0.79	0.85	0.88
	Recall	0.93	0.75	0.84	0.88
	F1-score	0.92	0.77	0.84	0.88
	Accuracy				0.88

a series of experiments to explore its performance across various configurations. The quantum circuit's complexity is investigated by adjusting the number of qubits from 5, 6, 7, and 8 to understand the impact of system size on the model's ability to capture and process information. Additionally, the depth of the circuit is explored with 1, 2, and 3 layers, aiming to assess how additional layers influence the model's expressivity and learning capacity. Recognizing the critical role of optimization in training neural networks, modal is also evaluated over a spectrum of optimizers, including RMSprop, RMSprop with momentum, Adam, and Adamax (University 2025; Kingma and Ba 2017; Ruder 2017). Each optimizer is selected to probe different aspects of the optimization landscape, with learning rates meticulously adjusted to harmonize with each optimizer's unique characteristics. This comprehensive examination aimed to examine the efficacy, robustness, and scalability of a hybrid neural network model.

4.3 Training and evaluation

In the experimental study, the dataset, which comprised a total of 458,913 unique customers, is split into a 60:40 ratio for training and test data randomly, leading to 275,347 training examples and 183,566 test examples. In order to manage the high computational demands and facilitate efficient training of the model, the dataset is partitioned into a total of 276 mini-batches, each containing 1,000 examples. This partitioning strategy optimized the training process by leveraging batch-based gradient descent methods and also ensured a balanced utilization of computational resources, thereby enhancing the overall effectiveness of the model in the training phase. The cross-entropy loss function between the predicted probabilities and the actual labels is used as the criterion for optimizing the model parameters. The training process is conducted over a series of epochs, extending to a total of 20 epochs until the loss function values stabilized, a critical indicator of the model reaching a plateau in its learning capability, thereby suggesting the attainment of



(a) Confusion Matrix for training dataset (b) Confusion Matrix for test dataset

Fig. 6 Confusion matrix for six-qubit PQC architecture

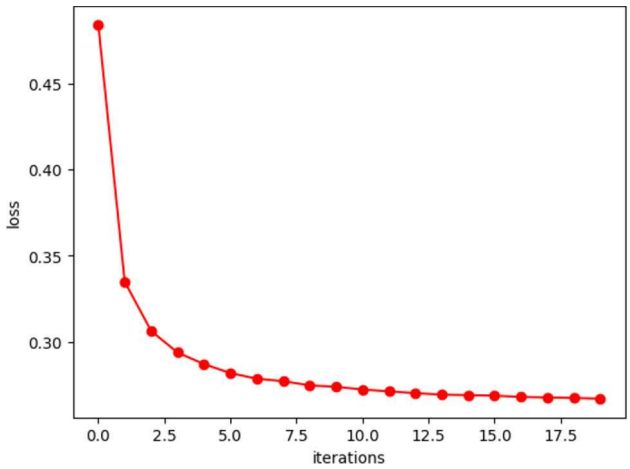


Fig. 7 Loss function for seven-qubit PQC architecture

an optimal balance between learning efficiency and model performance. Model performance is rigorously evaluated using a suite of metrics, including accuracy, precision, recall, and F1 score (Amex 2024b). To ensure the reliability of the findings, a consistent random seed is set across all experimental runs to minimize variability due to stochastic processes inherent in machine learning training, thereby enhancing the robustness and credibility of the results.

The training process involved the utilization of gradient-based optimization algorithms to minimize the loss function. Techniques such as dropout are employed to prevent overfitting and ensure the model's generalizability. The model's performance is evaluated using a confusion matrix appropriate for binary classification tasks. Apart from accuracy, matrices like precision, recall, and F1-Score provided a more nuanced view of the model's performance, especially in the presence of class imbalances.

Table 3 Performance metrics for the training and test datasets for seven-qubit PQC architecture

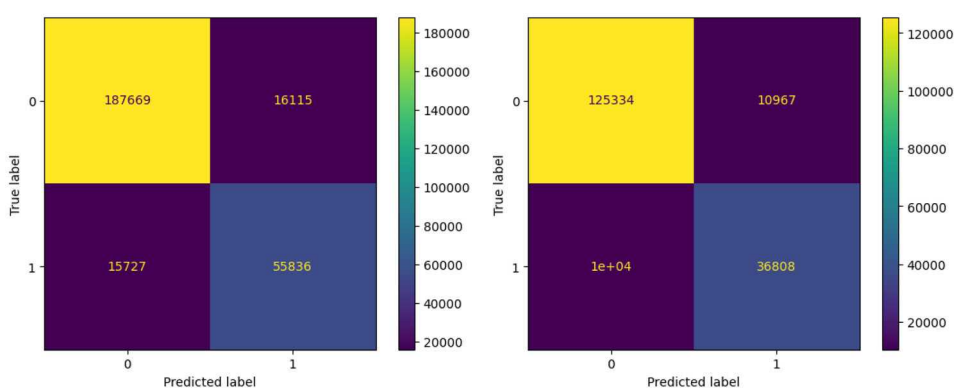
Dataset	Metric	Class 0	Class 1	Macro avg	Weighted avg
Training	Precision	0.92	0.78	0.85	0.88
	Recall	0.92	0.78	0.85	0.88
	F1-score	0.92	0.78	0.85	0.88
	Accuracy				0.88
Test	Precision	0.92	0.77	0.85	0.88
	Recall	0.92	0.78	0.85	0.88
	F1-score	0.92	0.77	0.85	0.88
	Accuracy				0.88

4.4 Case 1: assessing quantum model scalability through qubit variation

The quantum circuit's complexity and its capacity to encode and process information are fundamentally determined by the number of qubits. The experiments involved methodically varying the number of qubits, starting from a baseline configuration and incrementally increasing the count. This approach allowed us to assess how augmentations in the quantum system's size influence the model's ability to capture complex patterns and perform computational tasks with varying degrees of difficulty.

4.4.1 Five-qubit PQC architecture

In this experiment, a network architecture is designed with five qubits to explore the impact of system size on model performance. A quantum circuit with a single layer, encompassing five rotation angle parameters, optimized via the



(a) Confusion Matrix for training dataset (b) Confusion Matrix for test dataset

Fig. 8 Confusion matrix for seven-qubit PQC architecture

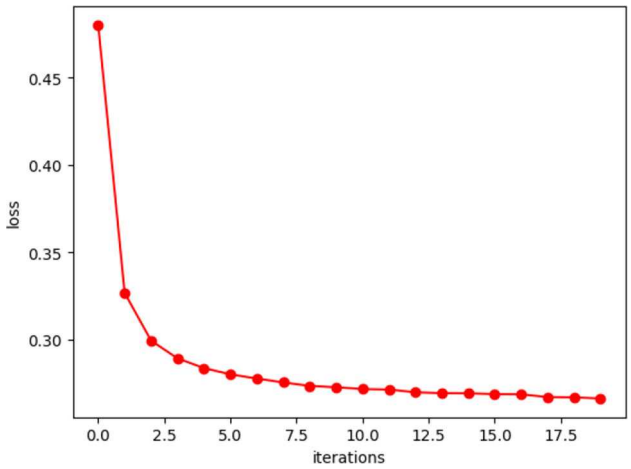


Fig. 9 Loss function for eight-qubit PQC architecture

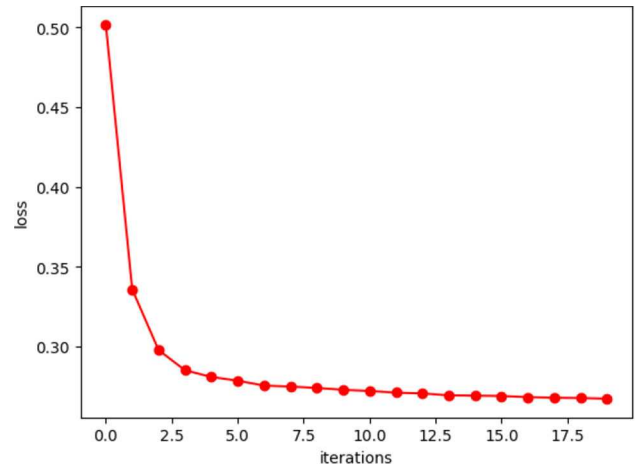


Fig. 11 Loss function for RMSprop algorithm

Adam optimizer-a first-order gradient-based optimization algorithm has been experimented with. The optimization parameters chosen for the experiment are as follows: step size $\alpha = 0.005$, exponential decay rates for the moment estimates $\beta_1 = 0.9$ and $\beta_2 = 0.999$, and a numerical stabilizer $\epsilon = 10^{-10}$.

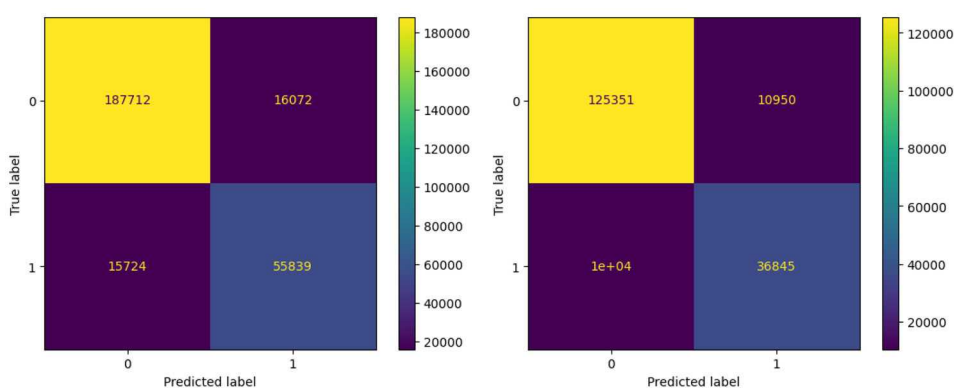
The iterative training process demonstrated a monotonic convergence, as evidenced by the loss function's trajectory, which plateaued at approximately 0.23296 after 20 epochs, indicating a stabilization in the model's learning (Fig. 3). This behavior of the loss function is emblematic of an effective optimization process, where the initial rapid descent represents a swift adaptation to the data, followed by incremental refinements as the model approaches an optimum.

The efficacy of the model is quantitatively assessed using confusion matrices for both the training and test datasets, as shown in Fig. 4a and b, respectively. The corresponding metrics scores derived from the confusion matrices are shown in Table 1.

4.4.2 Six-qubit PQC architecture

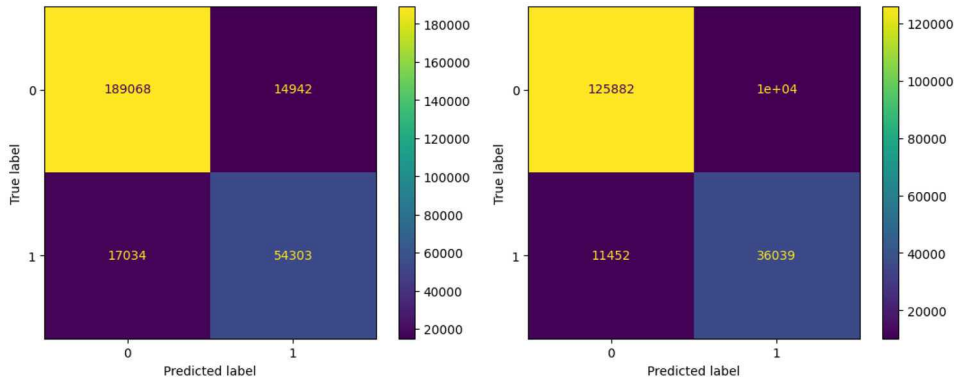
For this experiment, a network architecture is designed with six qubits to explore the impact of system size on model performance with the same parameters utilizing the Adam gradient as used in the previous experiment, a quantum circuit with a single layer, and encompassing six rotation angle parameters. The iterative training process in this experiment provided a minimum loss value of 0.23295 in 20 epochs (Fig. 5).

The confusion matrices for both the training and test datasets are shown in Fig. 6a and b, respectively. Their corresponding metrics scores are shown in Table 2.



(a) Confusion Matrix for training dataset (b) Confusion Matrix for test dataset

Fig. 10 Confusion matrix for eight-qubit PQC architecture



(a) Confusion Matrix for training dataset (b) Confusion Matrix for test dataset

Fig. 12 Confusion matrix for RMSprop algorithm

4.4.3 Seven-qubit PQC architecture

In this experiment, a network architecture with seven qubits, and a single layer, encompassing seven rotation angle parameters, is optimized via the Adam optimizer with the same parameters. The iterative training process demonstrated a monotonic convergence, as evidenced by the loss function's trajectory, which plateaued at approximately 0.2253 after 20 epochs (Fig. 7).

The confusion matrices for training and test datasets are shown in Fig. 8a and b, respectively. Their corresponding metrics scores derived from the confusion matrices are as shown in Table 3.

4.4.4 Eight-qubit PQC architecture

In this experiment, a PQC with eight qubits, and a single layer, encompassing eight rotation angle parameters, is optimized via the Adam optimizer with the same parameters. The iterative training process led to convergence after 20 epochs, providing a loss function value of 0.21719, indicating a stabilization in the model's learning (Fig. 9).

The confusion matrices for training and test datasets are shown in Fig. 10a and b, respectively. Their corresponding metric scores are shown in Table 4.

4.5 Case 2: evaluation of optimization algorithm effects on HNN model's convergence

Given the pivotal role of optimization algorithms in the training process of machine learning models, the experimental design included a comparative analysis of several optimization algorithms. This analysis encompassed a range of optimization algorithms such as RMSprop, RMSprop with momentum, Adam, and Adamax. Each optimizer is evaluated for its efficacy in navigating the model's parameter space towards optimal solutions, with a particular focus on its impact on convergence rates and overall model accuracy.

Table 4 Performance metrics for the training and test datasets for eight-qubit PQC architecture

Dataset	Metric	Class 0	Class 1	Macro avg	Weighted avg
Training	Precision	0.92	0.78	0.85	0.88
	Recall	0.92	0.78	0.85	0.88
	F1-score	0.92	0.78	0.85	0.88
	Accuracy				0.88
Test	Precision	0.92	0.77	0.85	0.88
	Recall	0.92	0.78	0.85	0.88
	F1-score	0.92	0.77	0.85	0.88
	Accuracy				0.88

Table 5 Performance metrics for the training and test datasets for RMSprop algorithm

Dataset	Metric	Class 0	Class 1	Macro avg	Weighted avg
Training	Precision	0.92	0.78	0.85	0.88
	Recall	0.93	0.76	0.84	0.88
	F1-score	0.92	0.77	0.85	0.88
	Accuracy				0.88
Test	Precision	0.92	0.78	0.85	0.88
	Recall	0.93	0.76	0.84	0.88
	F1-score	0.92	0.77	0.84	0.88
	Accuracy				0.88

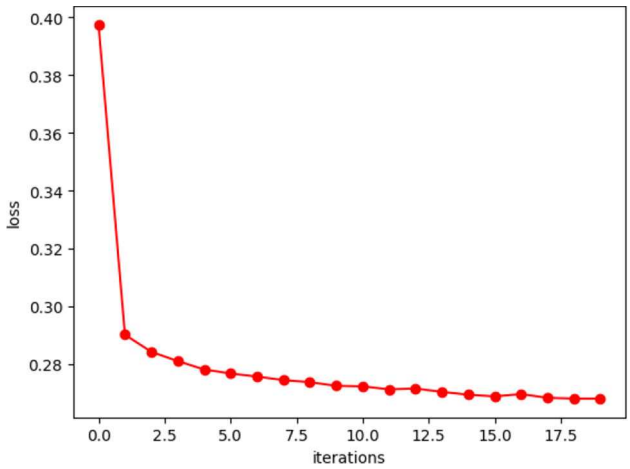


Fig. 13 Loss function for RMSProp with momentum algorithm

In Case 1, experiments are conducted employing the Adam optimizer, whereas subsequent investigations explored the efficacy of alternative optimization algorithms to ascertain their performance and suitability.

4.5.1 Optimization of PQC using the RMSprop algorithm

In this experiment, keeping the number of qubits as six, one layer of PQC with six rotation angles and utilizing the RMSprop optimizer with a step size of 0.005, gamma set to 0.9, and epsilon at 1×10^{-8} are chosen for the optimization and training of the model. During the iterative training process, the trajectory of the loss function stabilized at an approximate value of 0.21304 following the completion of 20 epochs, as depicted in Fig. 11. The confusion matrices for training and test datasets are shown in Fig. 12a and b, respectively. Their corresponding metrics scores are shown in Table 5.

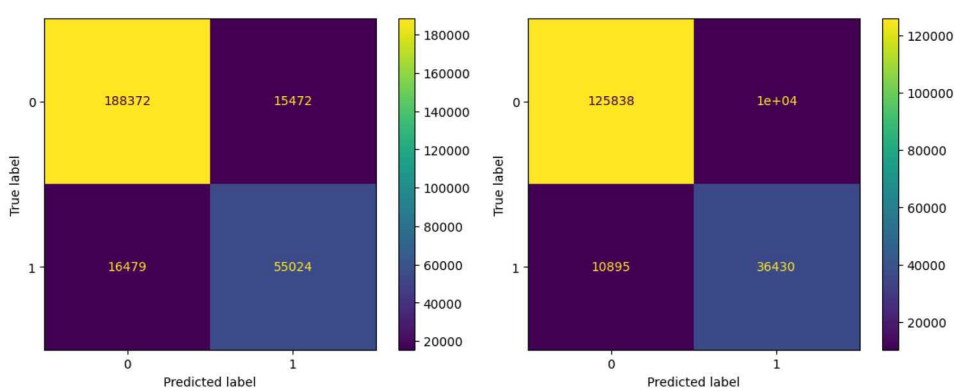
Table 6 Performance metrics for the training and test datasets for RMSProp with momentum algorithm

Dataset	Metric	Class 0	Class 1	Macro avg	Weighted avg
Training	Precision	0.92	0.78	0.85	0.88
	Recall	0.93	0.76	0.84	0.88
	F1-score	0.92	0.77	0.85	0.88
	Accuracy				0.88
Test	Precision	0.92	0.78	0.85	0.88
	Recall	0.93	0.76	0.84	0.88
	F1-score	0.92	0.77	0.84	0.88
	Accuracy				0.88

4.5.2 Optimization of PQC using the RMSProp with momentum algorithm

In this experiment, the network architecture is set up by utilizing six qubits and their corresponding rotation angle parameters, considering one layer of PQC, and optimizing using the RMSProp with momentum algorithm. RMSprop with momentum, a variant integrating momentum into the RMSprop algorithm, is utilized with a step size of 0.005, a gamma decay rate of 0.9, a stabilization constant epsilon of 1×10^{-8}, and a momentum factor of 0.6. This configuration aimed to enhance convergence speed and stability by adjusting the learning rate dynamically and incorporating momentum to navigate through the parameter space efficiently. During the iterative training process, the trajectory of the loss function stabilized at an approximate value of 0.217946 following the completion of 20 epochs, as depicted in Fig. 13.

The efficacy of the model is quantitatively assessed using confusion matrices for both the training and test datasets. The confusion matrices for training and test datasets are shown



(a) Confusion Matrix for training dataset (b) Confusion Matrix for test dataset

Fig. 14 Confusion matrix for RMSprop with momentum algorithm

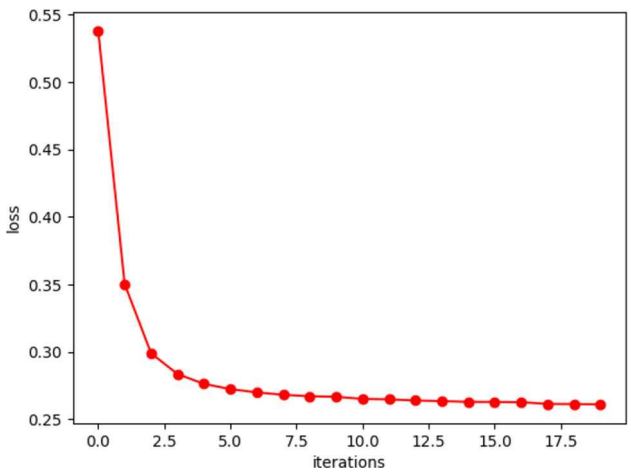


Fig. 15 Loss function for Adamax algorithm

in Fig. 14a and b, respectively. Their corresponding metric scores are shown in Table 6.

4.5.3 Optimization of PQC using the Adamax algorithm

In this experimental investigation, the Adamax optimizer, an extension of the Adam optimization algorithm, is employed. It is characterized by its utilization of the infinity norm for parameter updates. Configured with a step size of 0.005, a first-moment decay rate ($b1$) of 0.9, a second-moment decay rate ($b2$) of 0.999, and (ϵ) of 1×10^{-10}, this optimizer is meticulously chosen to enhance the convergence stability and efficiency of the model. During the iterative training process, the trajectory of the loss function stabilized at an approximate value of 0.22862 following the completion of 20 epochs, as depicted in Fig. 15. The confusion matrices for training and test datasets are shown in Fig. 16a and b, respectively. Their corresponding metric scores are shown in Table 7.

Table 7 Performance metrics for the training and test datasets for Adamax algorithm

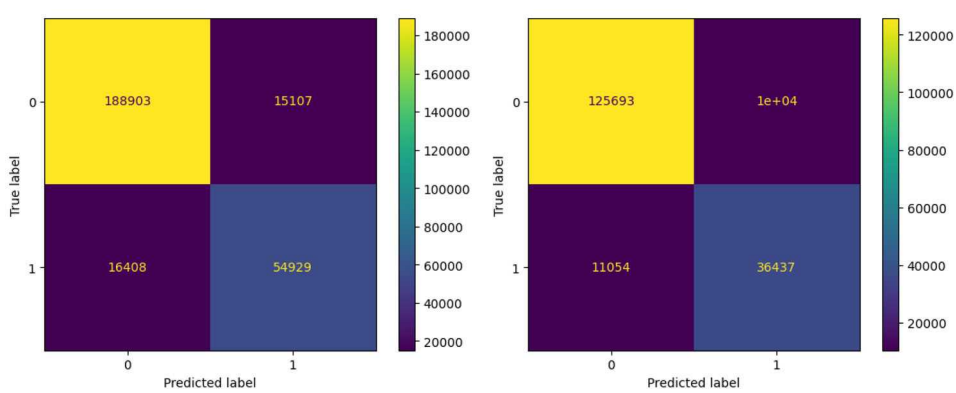
Dataset	Metric	Class 0	Class 1	Macro avg	Weighted avg
Training	Precision	0.92	0.78	0.85	0.88
	Recall	0.93	0.77	0.85	0.88
	F1-score	0.92	0.78	0.85	0.89
	Accuracy				0.89
Test	Precision	0.92	0.78	0.85	0.88
	Recall	0.92	0.77	0.85	0.88
	F1-score	0.92	0.77	0.85	0.88
	Accuracy				0.88

4.6 Case 3: quantum circuit depth and its influence on model expressivity and complexity

In cases 1 and 2, experiments have been conducted with one quantum layer, while case 3 will explore the effect of an increase in the depth of the quantum circuit on the HNN model. The depth of the quantum circuit, defined by the number of quantum layers, represents another critical dimension of the experimental investigation. The number of layers in a quantum circuit is indicative of the circuit's ability to perform complex transformations on the quantum state. The number of layers is increased to explore the trade-offs between model expressivity and the practical limitations imposed by the complexity of circuit design. This aspect of the study provided insights into the optimal circuit depth for balancing computational power and real-world feasibility.

4.6.1 Dual-layer PQC

In this experiment, a quantum machine learning model architecture is configured with six qubits and two quantum layers,



(a) Confusion Matrix for training dataset (b) Confusion Matrix for test dataset

Fig. 16 Confusion matrix for Adamax algorithm

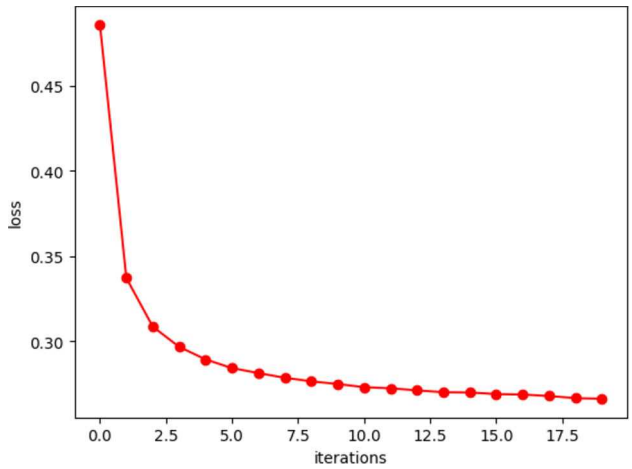


Fig. 17 Loss function for dual-layer PQC

Table 8 Performance metrics for the training and test datasets for dual-layer PQC

Dataset	Metric	Class 0	Class 1	Macro avg	Weighted avg
Training	Precision	0.92	0.77	0.85	0.88
	Recall	0.92	0.78	0.85	0.88
	F1-score	0.92	0.78	0.85	0.88
	Accuracy				0.88
Test	Precision	0.92	0.77	0.85	0.88
	Recall	0.92	0.76	0.85	0.88
	F1-score	0.92	0.77	0.85	0.88
	Accuracy				0.88

encompassing twelve rotation angle parameters, aiming to assess the impact of increased circuit complexity on computational performance. The Adam optimizer, retaining the parameters of a step size of 0.005, $b1 = 0.9$, $b2 = 0.999$, and (ϵ) of 1×10^{-10}, is applied to optimize the model, facilitating a nuanced exploration of the interplay between quantum circuit depth and learning efficacy. During the iterative training process, the trajectory of the loss function stabilized at an approximate value of 0.216275 following the completion of 20 epochs, as depicted in Fig. 17.

The efficacy of the model is quantitatively assessed using confusion matrices for both the training and test datasets, as shown in Fig. 18a and b, respectively. Their corresponding metric scores are shown in Table 8.

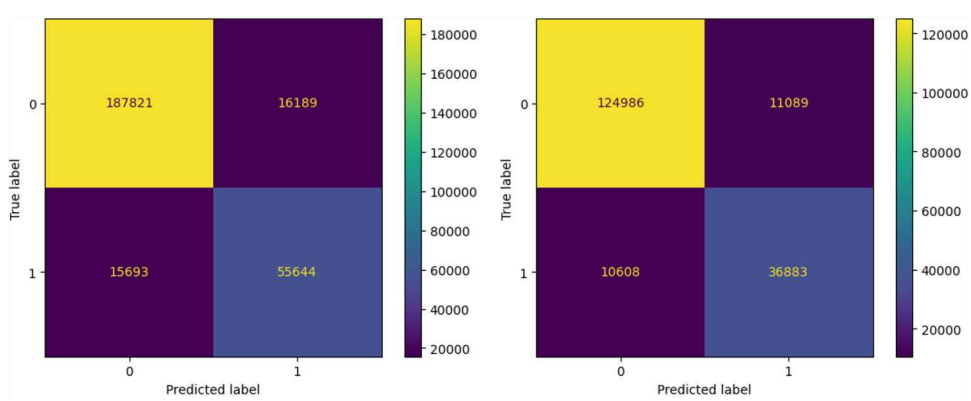
4.6.2 Triple-layer PQC

For this experiment, the quantum machine learning model architecture is configured with six qubits and three quantum layers, encompassing eighteen rotation angle parameters, aiming to assess the impact of further increased circuit complexity on computational performance. The Adam optimizer, retaining the previously established parameters of step size of 0.005, $b1$ of 0.9, $b2$ of 0.999, and (ϵ) of 1×10^{-10}, is trained over a span of 20 epochs to optimize the model, facilitating a nuanced exploration of the balance between expressivity and practical training efficiency within this enhanced quantum framework. During the iterative training process, the trajectory of the loss function stabilized at an approximate value of 0.215686 following the completion of 20 epochs, as depicted in Fig. 19.

The efficacy of the model is quantitatively assessed using confusion matrices for both the training and test datasets. The confusion matrices for the training and test datasets are shown in Fig. 20a and b, respectively. Their corresponding metric scores are shown in Table 9.

4.7 Speedup in training time

In recent years, the landscape of machine learning has been significantly enriched by the advent of advanced boosting



(a) Confusion Matrix for training dataset (b) Confusion Matrix for test dataset

Fig. 18 Confusion matrix for dual-layer PQC

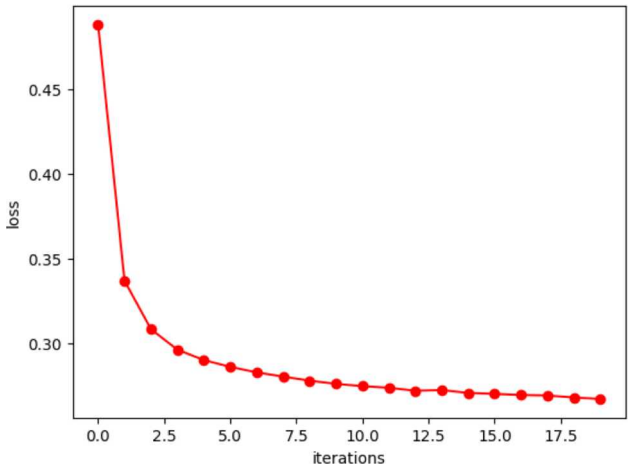


Fig. 19 Loss function for triple-layer PQC

Table 9 Performance metrics for the training and test datasets for triple-layer PQC

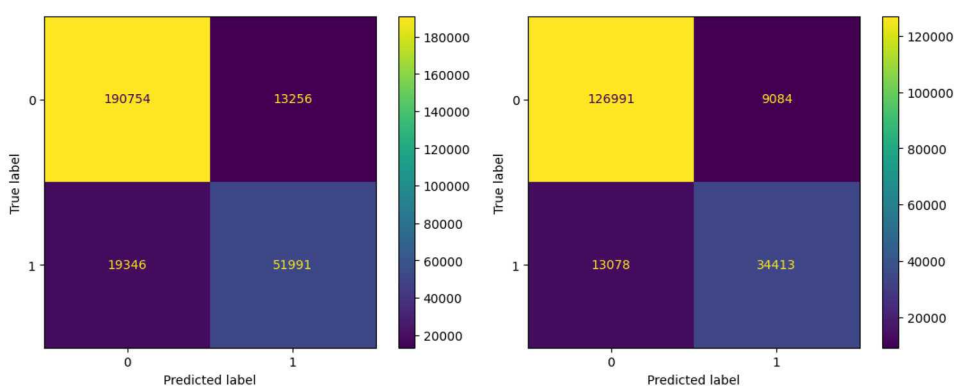
Dataset	Metric	Class 0	Class 1	Macro avg	Weighted avg
Training	Precision	0.91	0.80	0.85	0.88
	Recall	0.94	0.73	0.83	0.88
	F1-score	0.92	0.76	0.84	0.88
	Accuracy				0.88
Test	Precision	0.91	0.79	0.85	0.88
	Recall	0.93	0.72	0.83	0.88
	F1-score	0.92	0.76	0.84	0.88
	Accuracy				0.88

algorithms such as XGBoost, LightGBM, and CatBoost. These methodologies have emerged as powerful tools in predictive analytics. The investigation aims to conduct a comprehensive comparison between hybrid neural networks and state-of-the-art boosting algorithms training times over the time-series dataset. The efficiency of the model's training has become a pivotal concern, especially as datasets grow in size and complexity. This paper presents a detailed comparison between HNNs and established boosting algorithms like XGBoost (Bareti 2024), LightGBM (Skloveyyp 2024), and CatBoost (Otugrul 2024), specifically through the lens of training time efficiency as shown in Table 10. However, as computational resources and time become increasingly critical constraints, Hybrid Neural Networks, with their integration of classical and quantum computing or other advanced computational techniques, promise to revolutionize the machine learning landscape by potentially offering faster convergence and processing capabilities. Conversely, boosting algorithms like XGBoost, LightGBM, and CatBoost are widely recognized for their speed and efficiency in training. This comparative analysis provides a comprehensive overview of their operational efficiency in terms of their training time for a machine learning model.

By focusing on the training timings of HNNs versus the boosting algorithms, the study offers valuable insights into the computational speed offered by HNNs. This exploration is critical as it directly impacts the feasibility and scalability of machine learning solutions in real-world applications.

5 Discussion

To construct a resilient HNN with integrated PQC, a sophisticated PQC is meticulously developed and intricately tailored to adeptly navigate and encapsulate the multifaceted features and inherent high dimensionality characteristic of time-series financial datasets. This advanced architectural design is predicated on leveraging the unique computational paradigms offered by quantum mechanics, thereby enabling the model to efficiently process and analyze complex temporal patterns and relationships that are pivotal in financial forecasting and risk assessment tasks. A thorough investigation is carried out



(a) Confusion Matrix for training dataset (b) Confusion Matrix for test dataset

Fig. 20 Confusion matrix for triple-layer PQC

Table 10 Training time comparison

Model	Training time (s)
XGBoost	4991.3
LightGBM	3985.4
CatBoost	3884.2
HNN	453.3

Table 12 Loss values corresponding to depth of quantum circuit

Circuit depth	Loss
1	0.23295
2	0.21627
3	0.21568

to elucidate the impact of varying critical parameters within the model's framework, aiming to assess its robustness, performance, and scalability. This investigation specifically focused on three pivotal aspects: the number of qubits, which determines the computational space of the quantum circuit; the selection of optimization algorithms, crucial for the efficient convergence of the model; and the depth of the quantum circuit, indicative of the circuit's complexity.

The augmentation of qubits within the quantum circuit yielded a reduction in the value of the loss function and a marginal enhancement in the model's overall performance (Table 11). This increment suggests that expanding the quantum state space through additional qubits provides a richer computational substrate for modeling complex patterns inherent in the data. Similarly, an increase in the quantum layers also led to a reduction in the value of the loss function while maintaining consistency across performance metrics (Table 12). Such consistency underscores the robustness of the quantum model, affirming its capability to sustain predictive accuracy despite escalating circuit complexity. This attribute is indicative of the inherent scalability of quantum-enhanced models, positing them as viable contenders for addressing sophisticated computational challenges without precipitating a decline in performance efficacy. Among the various optimizers trialed, the RMS optimizer emerged as marginally superior, optimizing the model's parameters more effectively in comparison to its counterparts (Table 13). This finding aligns with the optimizer's theoretical underpinnings, which are designed to adaptively adjust learning rates, thereby facilitating efficient convergence. Notwithstanding these advancements, a notable increase in training duration and computational demands is observed concomitant with the expansion of qubits and quantum layers. This phenomenon accentuates the imperative for a judicious balance between the quantum circuit's complexity and the operational constraints, aiming to optimize the trade-off between model performance and computational efficiency. Therefore, calibration of quantum circuit parameters is necessary to harmonize the dual objectives of enhancing model performance while mitigating computational overheads.

A noteworthy accomplishment of the study is the speedup observed in the training duration of the HNN, surpassing the state-of-the-art gradient-boosting classical algorithms on the same time-series dataset (Table 10), which underscores the potential influence of such networks in managing the escalating volumes of data. The insights garnered from this study significantly contribute to the burgeoning discourse on quantum machine learning, paving the way for future explorations aimed at harnessing the full potential of quantum-enhanced computational models. This study introduces a robust and scalable hybrid machine learning model that leverages the synergistic advantages of both computing paradigms for predictive analytics. Specifically tailored for application on highly sophisticated, voluminous, and complex financial datasets, the model embodies a cutting-edge approach to navigating the intricacies of financial data. Through the integration of quantum computing's unparalleled computational capabilities with the mature, versatile tools of classical machine learning, this research delineates a novel pathway for enhancing predictive accuracy and efficiency in the analysis of intricate financial time-series data.

6 Conclusion

This study has meticulously explored the integration of classical and quantum computing with machine learning to address the challenges presented by sophisticated financial datasets characterized by their high dimensionality and

Table 11 Loss values corresponding to qubit count

Qubit count	Loss
5	0.23296
6	0.23295
7	0.22530
8	0.21719

Table 13 Loss values corresponding to optimization algorithm

Optimization algorithm	Loss
RMS	0.21304
RMS with momentum	0.21794
Adam	0.23295
Adamax	0.22862

complexity. The hybrid machine learning model presented in this study is predicated upon a novel and sophisticated state-of-the-art parametrized quantum circuit, meticulously developed to facilitate predictive financial analysis. This novel approach is specifically tailored to address the challenges posed by complex, voluminous, and high-dimensional time-series datasets. Through the strategic integration of quantum computing mechanisms with classical machine learning techniques, the model sets a new benchmark in the realm of financial analytics, offering unprecedented computational capabilities and insights into the intricate dynamics of financial markets. Through a series of methodically designed experiments, we have investigated the influence of key parameters—namely, the number of qubits, the choice of optimization algorithms, and the depth of quantum layers—on the model's performance, robustness, and scalability. The significant advantage manifested as a reduction in training time by the novel hybrid neural network, when compared to the state-of-the-art gradient-boosting classical algorithms, underscores its practical applicability and efficiency, particularly in real-world financial analytics scenarios accommodating the burgeoning volume of data. This study is a step towards innovative solutions and contributes to the burgeoning field of quantum machine learning, demonstrating the feasibility and advantages of hybrid classical-quantum approaches for predictive analytics in finance to harness the full potential of quantum computing in tackling the complexities of financial datasets. The speedup, scalability, and robustness demonstrated by the novel hybrid neural network architecture proposed in this study pave the way for its application across a broader spectrum of data-intensive domains, promising significant advancements in predictive analytics and beyond.

Author contribution Both authors contribute equally to the literature review, writing the text, and reviewing the manuscript.

Data availability No datasets were generated or analyzed during the current study.

Declarations

Conflict of interest The authors declare no competing interests.

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